

Trig with Ln and e

section 5.2

11/17/25

$$\int \frac{x^3 - 6x}{x^2 + 3} dx =$$

$$\begin{array}{r} x^2+3 \overline{) \begin{array}{r} x^3 + 0x^2 - 6x + 0 \\ -(x^3 + + 3x) \\ \hline -9x \end{array}} \end{array}$$

$$\int x - \frac{3x}{x^2+3} dx$$

$$u = x^2 + 3$$

$$du = 2dx$$

$$\int x - \int \frac{\frac{a}{2} \cdot \frac{1}{u}}{u} du$$

$$\frac{1}{2}x^2 - \frac{a}{2} \ln(x^2+3) + C$$

Trig

Rule 1

$$\int \tan x dx = \int \frac{\sin x}{\cos x} dx = -\int \frac{1}{u} du = -\ln|\cos x| + C$$

RULE 2

$$\int \cot(x) dx = \ln|\sin x| + C$$

Rule 3

$$\int \sec x dx \cdot \frac{\sec x + \tan x}{\sec x + \tan x} =$$

$$= \int \frac{\sec^2 x + \sec x \tan x}{\sec x \tan x} dx$$

$$= \ln|\sec x + \tan x| dx$$

Rule 4

$$\int \csc x dx = -\ln|\csc x + \cot x| + C$$

$$\int e^x \sec(e^x) dx =$$

$$u = e^x$$

$$du = e^x dx$$

$$\int \sec(u) du = \ln|\sec(e^x) + \tan(e^x)| + C$$

$$\int \frac{\tan(\ln y)}{y} dy =$$

$$u = \ln y$$

$$du = \frac{1}{y} dy$$

$$\int \tan(u) du$$